Strong generalization for the sequent calculus

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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel thm:strong-generalization of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- LK derivations and the right universal quantifier rule
- the eigenvariable condition on constants in R forall inferences
- finiteness of derivations giving a finite premise set Gamma_0

Theorem 0.1. *If c is a constant not occurring in Γ or $!A(x)$ and $\Gamma \vdash !A(c)$, then $\Gamma \vdash \forall[x][!A(x)]$.*

Proof. Let π_0 be an LLK-!!derivation of $\Gamma_0 \Rightarrow !A(c)$ for some finite $\Gamma_0 \subseteq \Gamma$. By adding a inference, we obtain !!aderivation of $\Gamma_0 \Rightarrow \forall[x][!A(x)]$, since c does not occur in Γ or $!A(x)$ and thus the eigenvariable condition is satisfied. □

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